

ROBIN BILODEAU

Ottawa, Ontario, Canada

rbilo073@uottawa.ca ♦ (+1) 613 371 1300 ♦ linkedin.com/in/robin-bilodeau ♦ github.com/RobinB1lo

EDUCATION

University of Ottawa

Ottawa, Ontario, Canada

Bachelor of Science in Computer Science and Mathematics

Sep 2024 – Apr 2028

- **Relevant Courses:** Intro to Computing 1, Intro to Computing 2, Linear Algebra, Differential Equations and Laplace Transforms, Discrete Structures, Elementary Real Analysis, Introduction to probability

WORK EXPERIENCE

Scotiabank - Global Markets

Toronto, Ontario, Canada

Quantitative Analyst Intern

Sep 2025 – Dec 2025

- Refactored a legacy pricing process into a modular architecture supporting multiple rates derivatives, including swaps, swaptions, caps, and floors
- Optimized performance bottlenecks in core analytical functions, leading to substantial runtime improvements
- Conducted in-depth testing and investigation of pricing models to identify product-dependent discrepancies and assess model validity, and visualized discrepancy differences across products to communicate findings clearly

RESEARCH EXPERIENCE

Privacy-Preserving Zero-Knowledge Credential Protocol

Ottawa, Ontario, Canada

Undergraduate Researcher

Feb 2026 – Present

- Implementing a zero-knowledge proof system over Pedersen commitments enabling privacy-preserving equality testing on hidden attributes, combining digital signatures, hashing, and authenticated encryption

A Data-Driven Assessment of Machine Learning Models for Climate-Sensitive Wildfire Susceptibility in Alberta

Ottawa, Ontario, Canada

Undergraduate Researcher

Feb 2026 – Present

- Developing and extending the GMDH algorithm to produce an interpretable, self-organizing machine learning pipeline for wildfire susceptibility detection
- Abstract accepted to Earth 2026 Conference

Uncertainty-Aware Statistical Extensions for HDBSCAN Clustering

Ottawa, Ontario, Canada

Undergraduate Researcher

Apr 2025 – Oct 2025

- Developed a Monte Carlo sampling framework to repeatedly adjust probability rank matrices, yielding statistically stable core-distance estimates for clustering high-noise data leading to reduced sensitivity to outliers and improved cluster detection reliability
- Coordinated a fast Fourier-based deconvolution estimator to recover underlying data distributions from noise-contaminated observations, enhancing core distance estimates for robust HDBSCAN clustering

LEADERSHIP AND ACTIVITIES

SSC Case Study Competition - Parkinson's Disease Classification

Hamilton, Ontario, Canada

Team Member

Apr 2026 – Jun 2026

- Performed end-to-end data cleaning on a high-dimensional clinical dataset (500+ variables) from the Canadian Open Parkinson Network (C-OPN), including leakage analysis to identify and remove variables that directly encode diagnostic criteria for PD vs. atypical parkinsonism, and outcome-stratified MICE imputation to handle missing data
- Developed a random forest classifier achieving greater than 95% accuracy in distinguishing Parkinson's disease from atypical parkinsonism syndromes, using stratified k-fold cross-validation with fold splitting prior to imputation

SKILLS, LANGUAGES, INTERESTS

- **Languages:** English (Native speaker), French (Native speaker)
- **Programming:** Python, R, Java, Typst, C++ (Beginner)
- **Tools:** Microsoft Word, Git, Visual Studio Code, Canva, Jupyter
- **Libraries:** matplotlib, numpy, Scikit-learn, pandas, yfinance
- **Athletics:** Boxing, Hockey, Soccer, Volleyball
- **Activities:** Tutoring, Mathematics Club, Undergraduate Research Club